FCSG-Net for Composite Image Restoration

[Official student name]



Department of Computer Science and Engineering
National Institute of Technology Rourkela

FCSG-Net for Composite Image Restoration

Project report submitted in partial fulfillment

of the requirements for the degree of

Bachelor of Technology

in

Computer Science and Engineering

by

[Official student name]

(Roll Number: [Roll number])

based on research carried out

under the supervision of

Prof. [Supervisor name]

[Month], [Year]

Department of Computer Science and Engineering
National Institute of Technology Rourkela



Department of Computer Science and Engineering
National Institute of Technology Rourkela

[Month] [Day], [Year]

Certificate of Examination

Roll Number: *[Roll number]*
Name: *[Official student name]*
Title of Dissertation: *FCSG-Net for Composite Image Restoration*

We the below signed, after checking the project report mentioned above and the official record book (s) of the student, hereby state our approval of the project report submitted in partial fulfillment of the requirements of the degree of *Bachelor of Technology* in *Computer Science and Engineering* at *National Institute of Technology Rourkela*. We are satisfied with the volume, quality, correctness, and originality of the work.

[Supervisor name] Principal Supervisor	[Committee member 1] Member, DSC
[Committee member 2] Member, DSC	[Committee member 3] Member, DSC
[External examiner] External Examiner	[Chairperson] Chairperson, DSC

[Head of department]
Head of the Department

Department of Computer Science and Engineering
National Institute of Technology Rourkela

Prof. [Supervisor name]
Professor

[Month] [Day], [Year]

Supervisor's Certificate

This is to certify that the work presented in the project report entitled *FCSG-Net for Composite Image Restoration* submitted by *[Official student name]*, Roll Number [Roll number], is a record of original research carried out by him under my supervision and guidance in partial fulfillment of the requirements of the degree of *Bachelor of Technology* in *Computer Science and Engineering*. Neither this project report nor any part of it has been submitted earlier for any degree or diploma to any institute or university in India or abroad.

_____
[Supervisor name]

Declaration of Originality

I, *[Official student name]*, Roll Number *[Roll number]* hereby declare that this project report entitled *FCSG-Net for Composite Image Restoration* presents my original work carried out as a undergraduate student of NIT Rourkela and, to the best of my knowledge, contains no material previously published or written by another person, nor any material presented by me for the award of any degree or diploma of NIT Rourkela or any other institution. Any contribution made to this research by others, with whom I have worked at NIT Rourkela or elsewhere, is explicitly acknowledged in the dissertation. Works of other authors cited in this dissertation have been duly acknowledged under the sections "Reference" or "Bibliography". I have also submitted my original research records to the scrutiny committee for evaluation of my dissertation.

I am fully aware that in case of any non-compliance detected in future, the Senate of NIT Rourkela may withdraw the degree awarded to me on the basis of the present dissertation.

[Month] [Day], [Year]
NIT Rourkela

[Official student name]

Abstract

[Write the abstract last. State the problem, proposed approach, evaluation scope, and current status. Use completed results only when verified. Maximum two pages.]

Keywords: [Add three to seven keywords or phrases.]

Contents

Certificate of Examination ii

Supervisor's Certificate iii

Declaration of Originality iv

Abstract v

1 Project overview 1
- 1.1 Problem and objective . . . 1
- 1.2 What we are doing . . . 1
- 1.3 Scope . . . 1

2 Proposed pipeline 2
- 2.1 Pipeline overview . . . 2
- 2.2 Pipeline components . . . 2
- 2.3 Training objective . . . 2

3 Evaluation plan and current status 3
- 3.1 Data and comparison protocol . . . 3
- 3.2 Evaluation measures . . . 3
- 3.3 Completed and pending work . . . 3
- 3.4 Limitations and next steps . . . 3

References 4

Chapter 1

Project overview

1.1 Problem and objective

[Explain the restoration problem, its motivation, and the project objective in one short paragraph.]

1.2 What we are doing

[Describe the implemented approach and planned comparisons. Separate completed work from proposed work, and cite reused methods.]

1.3 Scope

[State the dataset and evaluation scope. List the questions this project can answer and the limits of the evidence.]

Chapter 2

Proposed pipeline

2.1 Pipeline overview

[Insert one architecture diagram. Show the inference path, the input residual connection, and a separate training-only target and loss branch.]

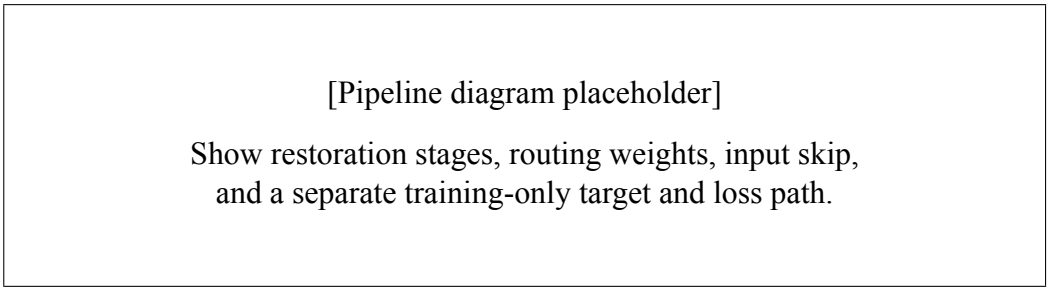


Figure 2.1: Proposed restoration pipeline. Replace this placeholder with the verified architecture diagram.

2.2 Pipeline components

[Describe each component briefly. Cite the source of reused ideas.]

Table 2.1: Pipeline component description template.

Component	Details to supply
Input and degradation	Image range, degradation order, and sampled ranges.
Frequency bands	Transform, complementary masks, and cutoffs.
Shared experts	Kernel sizes, blocks, channels, and pixel rearrangement.
Routing	Gate inputs, top-2 weights, and expert execution.
Fusion and output	Channel fusion, refinement, and input residual.

2.3 Training objective

[State the reconstruction, frequency, and routing terms used by the implementation. Define symbols and coefficients. Keep this to one short explanation.]

Chapter 3

Evaluation plan and current status

3.1 Data and comparison protocol

[Identify training and validation splits, synthetic degradations, DnCNN and FFDNet baselines, and the evaluation protocol. Separate crop validation from whole-image tiled evaluation.]

3.2 Evaluation measures

[List restoration metrics, parameter count, reported FLOPs, and measured latency. State hardware and measurement conditions. Do not equate lower FLOPs with faster inference.]

3.3 Completed and pending work

Table 3.1: Project status template. Replace each entry with verified information.

Work item	Status	Evidence or next action
[Implementation]	[Confirm]	[Source or validation record]
[Baseline comparison]	[Confirm]	[Completed evaluation files]
[Routing analysis]	[Confirm]	[Diagnostic evidence]
[Trained ablation]	[Confirm]	[Final evidence or pending action]

3.4 Limitations and next steps

[State the limits of the evidence and the next experiment. Include a small results table only if required for this submission. Do not turn historical progress logs into final results.]

References

[Add verified bibliographic records for sources cited in the report. Use one consistent style. This page is intentionally a template, with no invented references.]

Sources to verify before writing:

- Original DnCNN and FFDNet papers for the comparison methods.
- Frequency-restoration papers used to motivate the pipeline.
- Expert-routing and squeeze-and-excitation papers for reused concepts.
- Dataset and metric papers for the evaluation protocol.